

IB Computer Science

25 March 2026

Criteria A

Defining the problem:

The client (myself) wants to create an open source software program that is capable of loading various types of models with efficient loading techniques and a real time graphics environment to give an interesting and inclusive experience with the 3D models of their choice. However, creating a complete model loading pipeline from scratch is complex and difficult for many people to understand.

3D is a tough topic on its own, and that's not even including all the textures, meshes and node systems that models are packed full of. Although there are some libraries like Assimp, which makes these topics significantly more easy to understand and implement. But it's easy to watch tutorials and copy and paste code that you don't actually understand.

Rationale for the proposed solution.

The best idea for this was a stand alone desktop application because it gave the easiest access to things like the GPU, which is necessary for model loading. And ensures the program can run large models efficiently, without being limited by the internet or other browser restrictions like many web based solutions.

By developing this, the programmer (myself), will deepen their knowledge and understanding of computer graphic content, and concepts such as data structures for 3D geometry, rendering pipelines, and memory management between the CPU and GPU. Even though things like Unity and Babylon exist that make these kinds of things super easy, it would show that I truly understand many of the complex topics involved with model loading.

I also believe that having it be developed with C++ and OpenGL, creates a great mix of complexity and power, without having to deal with something like Vulkan or DX12.

Additionally, many resources like Assimp, will make many of the redundant model loading parts a lot easier and turn potentially frustrating parts into more enjoyable plug-and-play like experience, all the while teaching me the same concepts.

Success Criteria:

- 1) The application successfully loads and renders 3D models (.dae, .obj, .fbx, .GLTF, .GLB, etc), correctly rendered.
- 2) The system correctly processes the model into meshes, including all the vertices, indices, and texture coordinates.
- 3) The application supports texture mapping, ensuring models display with accurate materials.
- 4) The program demonstrates a clear model loading pipeline, including file importing, node traversal, mesh processing, and rendering.
- 5) The application loads textures efficiently, and correctly.
- 6) The program renders models at an interactive frame rate

- 7) Making it modular so that other things can be added later
- 8) The application provides visible evidence of correct functionality through multiple test models.